

Capture Fidelity

How a €30 webcam is made to measure a spectrum — and why every step is the way it is.

Audience. Written for three readers at once: the **developer** returning to this code after a month and needing the big picture; the **chemist** who wants to know whether the numbers can be trusted and what they physically mean; and the **lab** receiving a Spectracs report and asking how the instrument works. No code knowledge is required for chapters 1, 2 and 4 — code and file names appear `like this` and can be skipped.

What this is not. This is not a specification and it asks for no work. It is the consolidated record of what was measured, decided, and built. Every claim here is traceable to a spec section listed in Appendix B.

How to read it. Chapter 1 is a standalone summary — the abstract plus a one-table overview of every decision, for a reader who wants the impression without the argument. Chapters 2 and 3 then tell the story properly. Wherever a topic has more behind it than the story needs, a green box points into **Appendix A — Knowledge base**, which treats that topic properly and stands on its own. So the main text stays readable start to finish, and the background is there when you want it — including the brightness law and the sRGB standard, in KB 1.

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1. Abstract and summary

Chapter 1 stands alone. Read only this and you will know what the instrument is, what was decided, and why — without the derivations.

1.1 Abstract

The problem. Determine the quality of pressed pumpkin-seed oil — is it the prized deep green, or has over-roasting turned it brown? — using hardware a mill owner can afford: a diffraction grating, a white LED and a **consumer USB webcam**. A webcam is built to make pictures look good, not to measure light. It applies automatic exposure and white balance, interpolates colour it never measured, bends the brightness scale on purpose, and rounds everything to 8 bits. None of that is compatible with quantitative measurement.

The approach. Nothing is measured in absolute terms. Every measurement is a **ratio** of two spectra taken seconds apart — the pure solvent (R) and the oil dilution (S) — and the quantity that matters is a further ratio, of two absorbance bands. That double-ratio structure is what makes cheap hardware viable: the lamp's spectrum, the sensor's sensitivity, the grating, the lens, the exposure, the dilution and even the camera's brightness curve are all *multiplicative* factors that cancel. What is left over is a short, well-understood list — and this document is largely about that list.

The method. One rule shaped the work: **measure first, then decide**. Every chapter in §3 follows the same arc — a symptom, a measurement that explains it, a decision, and its cost. Several of them end in "change nothing", which is a legitimate outcome: the dark-frame chapter (§3.10) is a full investigation whose conclusion is that the textbook correction is not needed here, and it is precisely *because* it was measured that we can now rely on that.

The outcome. The instrument separates green from brown oils with a class separation of about **10× the run-to-run scatter**, and does so **independently of dilution** — a sloppy sample preparation cannot flip the verdict. Two errors were found and removed that had been biasing the reference (a systematically dim group of frames after each exposure change, and automatic white balance drifting between the two captures). One known error remains uncorrected: **sensor self-heating**, worth about 1.7 % of the reference's shape (§3.6). The full list of open items is in §6.

The surprises worth knowing about. Three findings ran against intuition and are documented in detail because a future reader will otherwise re-derive them the hard way:

- the camera's **exposure control runs backwards** and saturates — a higher setting gives a *darker* image, and above a threshold nothing changes at all (§3.4);
- ****undoing the camera's brightness curve properly**, with the official sRGB formula, makes the measurement worse** than using the simplified textbook version — by 24 % of the class separation (§3.3);
- the sensible-sounding fix of **moving the analysis band away from the noisy dark region** costs a quarter of the discriminating power, because the darkest part of the band *is* the signal. The real fix was to dilute the sample instead (§3.12).

1.2 What is actually being measured

A drop of pumpkin-seed oil dissolved in isopropanol is placed in the light path. The instrument never measures "how much light the oil absorbs" directly — it measures **two** spectra and divides them:

R(λ)	REFERENCE	pure isopropanol in the pot	-- "how much light arrives with no oil"
S(λ)	SAMPLE	the oil dilution in the pot	-- "how much light arrives through the oil"
$T(\lambda) = S(\lambda) / R(\lambda)$		transmission	– the fraction that got through
$A(\lambda) = -\log_{10}(T(\lambda))$		absorbance	– the Beer-Lambert quantity, ∝ concentration

Everything in this document exists to make that division trustworthy. **The ratio is the whole design idea**, and understanding what it does and does not cancel (§2.4) explains almost every decision that follows.

→ Background: **KB 2 — The Beer-Lambert law, and the algebra of why ratios are invariant**

1.3 The chain, end to end

LAMP	Yuji SunWave 6500 K white LED
	▼ light through the sample pot, through a slit, onto a diffraction grating
CAMERA	ELP UVC webcam @ 2592 × 1944, fixed exposure, fixed white balance
	(inside the camera: sensor → white balance → demosaic → GAMMA ENCODE → 8-bit)
	▼
FRAME	a 2-D image: horizontal axis = wavelength, vertical axis = the slit's height
	├ 1. crop to the region of interest (the bright band), drop the outer 20 % of rows
	├ 2. per pixel: grey = max(R, G, B) §3.2
	├ 3. per wavelength column: Tukey biweight down the rows (spatial) §3.7
	├ 4. repeat for 150 frames; reject dim frames, then sigma-clipped mean §3.8
	▼
SPECTRUM	one intensity value per wavelength, 440 ... 630 nm
	▼ done once for the reference, once for the sample
T = S / R → A = -log10(T) → evaluation (pigment bands, colour, verdict)	

1.4 The decisions at a glance

The whole of chapter 3 in one table: what the capture chain does, and what led us to do it that way. Each row is a section there.

§	What we do	What led us there
3.1	Pin capture to 2592 × 1944 and trip an alarm if the calibration does not fit the frame	Capture had silently drifted to a different resolution than the wavelength calibration was authored at. Every wavelength was wrong, and the spectrum still looked perfectly plausible.
3.2	Reduce each pixel to max(R, G, B)	The broadcast luminance formula it replaced weights blue at 5/32 — it was discarding two thirds of the blue signal, exactly where the pigment band sits. The camera image of that region was vividly blue; the <i>weighting</i> was suppressing it.

§	What we do	What led us there
3.3	Undo the camera's brightness curve with the pure $\gamma^{2.2}$ power law — not the official piecewise sRGB curve (<i>designed, not yet built</i>)	Absorbances are compressed by $\approx 1/2.2$. Measured: a pure power law leaves the verdict bit-identical at any exponent, while the physically more faithful sRGB curve costs 24 % of the class separation and buys no colour improvement.
3.4	Auto-expose with a direction-agnostic sweep on the 99.9th percentile of the per-channel maximum, then lock the exposure	The camera's exposure control is <i>inverted</i> and clamps, so the original bisection drove to the worst possible setting. A luminance metric hid a fully clipped green channel behind an unclipped red one.
3.5	Settle adaptively — drain frames until the frame brightness stops changing	After an exposure change the camera <i>ramps</i> for about a second. A fixed wait was a per-camera magic number, and an earlier adaptive test watched a quantity that plateaus early and declared success too soon.
3.6	Nothing yet — diagnosed and documented	The camera heats itself: the channel balance drifts as a clean exponential, $\tau = 171$ s, 1.7 % total, ~ 9 min to settle. The reference and sample straddle the steepest part of that curve. The largest error still uncorrected.
3.7	Reduce each wavelength column with a Tukey biweight over the central 60 % of rows	A single centre row let one hot pixel or dust speck corrupt a wavelength permanently — and a plain average over the rows is dragged by exactly the same pixel.
3.8	Reject dim frames as whole frames , then take a sigma-clipped mean	The reference burst runs right after an exposure sweep, so its first frames were systematically dim. A per-wavelength clip cannot reject a <i>coherent group</i> — the group inflates the spread and drags the mean toward itself.
3.9	Freeze white balance at 6500 K for measurement; leave it automatic for calibration	Automatic white balance re-converges after every exposure change, so the reference caught it mid-drift. It rescales channels <i>individually</i> , which is the one kind of change a ratio cannot cancel. Calibration needs the opposite, because it identifies emission lines by colour.
3.10	No dark-frame subtraction, no bad-pixel map	Measured over 150 dark frames at the worst-case exposure: black level 0.00 % of full scale. The offset the textbook correction removes is not there, and the ~ 10 hot pixels are already handled by §3.7.
3.11	Clamp the captured range to 440–630 nm , declared by the evaluation plugin	Outside that window the lamp is weak and the small optics roll off, so those wavelengths contribute noise dressed as data. The science declares what it needs; the capture layer obeys and fails loudly on a mismatch.
3.12	Prepare the sample at 1:30 or 1:33 , not 1:20	The fresh 2026 oils absorb far more than the aged ones used for validation: the sample bottomed out at DN 5 of 255 , inside the least trustworthy part of the camera's response. Diluting lifts it to 16–25 DN and changes the verdict by ± 0.35 %.
3.13	Rebuild the jar seat and mark the fill line; document the sample's own turbidity as the next lever	The jar is lifted out and put back between the two captures a ratio is made of, and any change in how it sits enters absorbance as an offset <i>and</i> a slope. Rebuilding the seat took re-seat tilt from 2.84 % to 1.34 % and metric scatter from 9.7 % to 2.9 % — the largest fidelity gain in the project, and a printed part rather than an algorithm. It also handed the top of the error budget to the chemistry: a few drops of oil in isopropanol is not a solution but a dispersion, and the scatter it adds is 0.7–1.9× the pigment signal .

Chapter 4 is the mirror image of this table — the things that were seriously considered and **deliberately not done**, with the reason for each.

1.5 The five things worth remembering

- 1. The reference cancels the instrument.** The lamp's spectrum, the sensor's per-wavelength sensitivity, the grating's efficiency and the lens's vignetting all appear identically in R and S , so they divide out exactly. This is why a cheap camera can do quantitative work at all, and why we do *not* need to characterise the sensor.
- 2. It cancels multiplicative errors, not additive ones.** Anything that *scales* the light cancels. Anything that *adds* to it — stray light, dark current, turbidity in the sample — survives the division and biases the result. **This is not a footnote: the sample's own turbidity adds a floor 0.7–1.9× the size of the pigment signal, and it is now the largest known error left in the instrument (§3.13).**
- 3. It cannot fix non-linearity.** The camera bends brightness on purpose (the "gamma" law). A ratio of two bent numbers is not the ratio of the true numbers. This is the one instrument error the reference leaves behind (§2.2, §3.3).
- 4. Ratios of two absorbance bands are immune to almost everything.** Dilution, gamma exponent, path length and overall gain all scale absorbance uniformly, and a ratio divides a uniform scale out. That is why the pumpkin verdict survives conditions that would wreck an absolute measurement.
- 5. Outliers come in two flavours and need two different cures.** A hot pixel is at the same place in every frame — only a spatial estimator kills it. A glitched frame is transient — only a temporal estimator kills it. Neither substitutes for the other (§2.5).

2. Foundations

Reference material. If you already know what an EOTF is, skip to chapter 3.

2.1 What a webcam actually hands you

A scientific (machine-vision) camera hands you a number proportional to the number of photons that hit the pixel. A **consumer webcam does not**. Between the sensor and the USB cable sits an image signal processor that has one job: make a picture that looks right on a screen. In order it applies

1. **automatic exposure** — adjusts integration time to a target brightness,
2. **automatic white balance** — rescales the R, G, B channels so white objects look white,
3. **demosaicing** — interpolates the colour filter mosaic into full-colour pixels,
4. **gamma encoding** — bends the brightness scale (the subject of §2.2),
5. **quantisation** — rounds to 8 bits, i.e. integers 0...255.

Only step 5 is unavoidable. Steps 1 and 2 are *turned off or frozen* by Spectracs (§3.4, §3.9) because a control loop that changes between the reference and the sample destroys the ratio. Step 4 cannot be turned off on a UVC webcam — it can only be undone in software (§3.3).

→ Background: **KB 5 — How a colour camera sees light: Bayer mosaic, demosaicing, quantum efficiency**

Throughout this document a stored 8-bit value is called a **DN** (*digital number*), 0...255. A DN is not a physical quantity; it is a byte.

→ Background: **KB 4 — Digital numbers, quantisation and noise**

2.2 The brightness law (gamma) — why it exists at all

Human vision is far more sensitive to differences among dark tones than among bright ones. If you store brightness *linearly* in 8 bits, the dark half of the range gets too few code values and shows visible banding, while the bright half gets more than the eye can use.

So every consumer imaging standard stores a **bent** version of brightness:

$$\begin{array}{lll} \text{stored value} & = & \text{light} ^ { (1 / \gamma) } \qquad \text{with } \gamma \approx 2.2 \qquad \text{(encoding, in the camera)} \\ \text{light} & = & \text{stored value} ^ \gamma \qquad \qquad \qquad \text{(decoding, in the display)} \end{array}$$

The bend allocates more code values to the dark end, where the eye needs them. Your monitor applies the inverse bend, so the picture looks right. The chain is invisible **as long as you only look at the picture**. It becomes a problem the moment you do arithmetic on the stored values — as we do.

The consequence for us. With $v = v_{\text{linear}}^{(1/\gamma)}$: $T_{\text{measured}} = S/R = (S_{\text{lin}}/R_{\text{lin}})^{(1/\gamma)} = T_{\text{true}}^{(1/\gamma)}$, hence $A_{\text{measured}} = A_{\text{true}} / \gamma$. Our absorbances are compressed by a factor of about 2.2 — they are real, but not *physically* real. And because that factor is **uniform**, any *ratio* of two absorbance bands is unaffected.

→ Background — the full treatment, both formulas, the terminology (OETF/EOTF), the history and the classic pitfalls: **KB 1 — The brightness law (gamma) in depth**

2.3 The variants — pure power law versus the sRGB standard

Two different curves both get called "the 2.2 law", and the difference matters at the dark end.

Variant A — the pure power law. One clean formula, `light = (DN/255)^2.2` . It is the textbook description and the one everybody quotes.

Variant B — the sRGB standard (IEC 61966-2-1, 1996), which is what cameras and monitors actually implement. It is **piecewise** — defined in two branches with a switchover point (the *knee*):

DN/255 ≤ 0.04045 → light = (DN/255) / 12.92 ← a straight line

DN/255 > 0.04045 → light = ((DN/255 + 0.055) / 1.055) ^ 2.4 ← a power curve

Two surprises live in that formula:

- **The exponent says 2.4, not 2.2.** The 2.4 is a *construction parameter* of the upper branch only. Together with the `+0.055 / 1.055` offset and the straight lower branch, the curve as a whole behaves like an exponent of ≈ 2.2 . Measured on our own oil spectra, the effective exponent comes out **2.10 – 2.24** across the mid-range — nowhere near 2.4. *2.2 is the answer; 2.4 is an ingredient.*
- **The straight line at the bottom is deliberate.** A pure power curve has *infinite slope at zero* in the encoding direction: the darkest code values would represent vanishingly small light differences, wasting codes and hugely amplifying noise when decoded. The standard caps the slope at a finite 12.92 and splices a straight line below the knee. That region — roughly **DN ≤ 10** — is called the **toe**.

The two variants agree closely everywhere above the knee and disagree noticeably below it. Which one to use is therefore a question about *how dark your darkest measurement is*. For Spectracs the answer turned out to be counter-intuitive and is the subject of §3.3.

→ Background: **KB 1 — the exact sRGB formulas, and where the two variants diverge**

2.4 What a reference measurement cancels — and what it does not

This is the single most useful mental model in the whole system. Write what the camera reports as

measured = g(λ) · true(λ) + D

then bent by the gamma law

where `g(λ)` collects **everything multiplicative** — lamp brightness at that wavelength, grating efficiency, lens transmission, sensor quantum efficiency, exposure time, gain — and `D` collects **everything additive** — sensor dark current, black-level offset, stray light inside the housing.

effect	multiplicative or additive?	does <code>T = S/R</code> remove it?
lamp spectrum (the LED's own shape)	multiplicative	yes, exactly
sensor spectral response (QE)	multiplicative	yes, exactly
grating efficiency, lens vignetting	multiplicative	yes, exactly
exposure time, analogue gain	multiplicative	yes — if identical in R and S
dark current, black level, stray light	additive	no — survives, biases T toward 1
turbidity (a cloudy, badly dissolved sample)	additive	no — looks like absorption
gamma encoding	neither — non-linear	no — leaves a uniform <code>1/γ</code> on absorbance

Read the table twice. It explains, in one place:

- why we never measured the sensor's spectral response, and never will (§4);
- why the dark level had to be *measured* before we could ignore it (§3.10);
- why "dissolve it properly" is the only lab instruction that really matters (§3.12);
- why exposure must be **locked** between reference and sample (§3.4);
- why gamma is the one residual worth a chapter of its own (§3.3).

→ Background: **KB 2 — the algebra of the ratio, and exactly where it stops holding**

2.5 Two kinds of outlier need two different cures

A wrong pixel value can come from two completely different places, and no single estimator removes both:

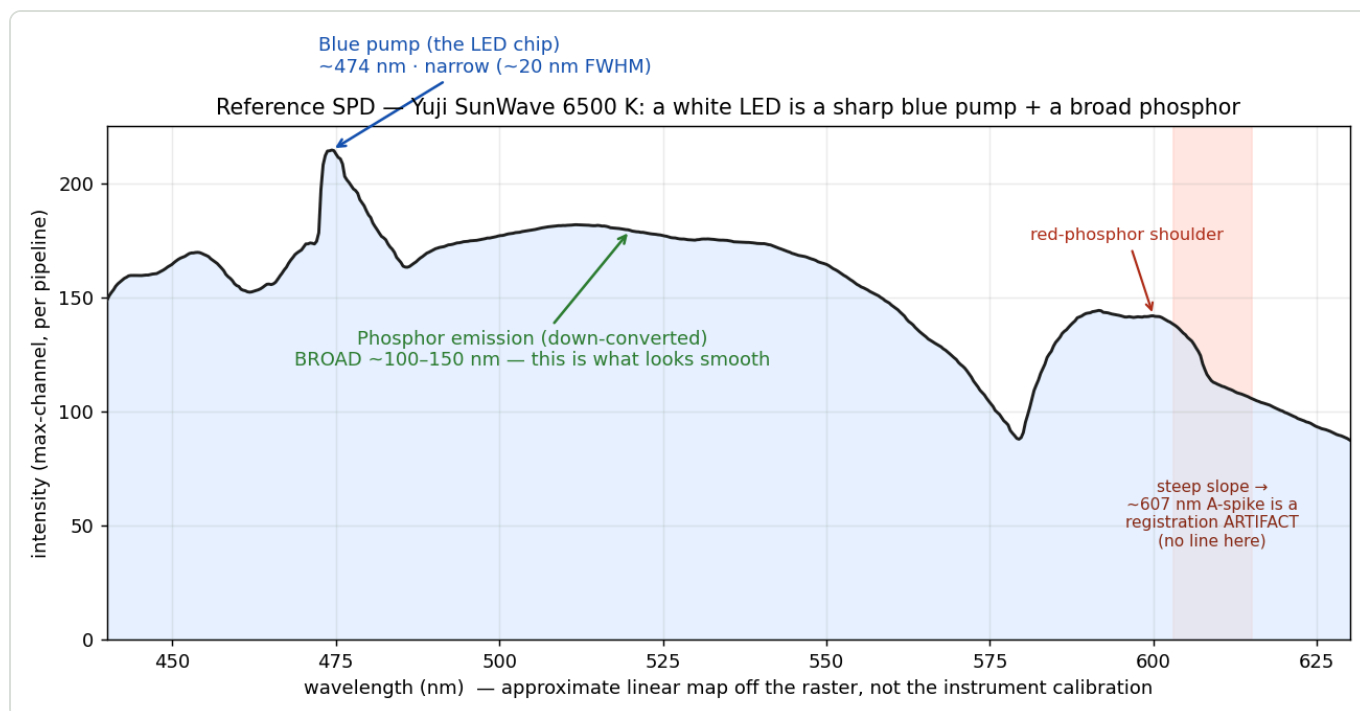
	where it comes from	it appears...	the only cure
spatial	a hot / dead pixel, a dust speck, a smile-distorted edge row	in the same place in every frame	a robust estimator across rows
temporal	a USB glitch, a read spike, an exposure ramp	in one frame, at all places	a robust estimator across frames

Averaging 150 frames does nothing at all to a hot pixel — it is in all 150. Averaging down the rows does nothing to a glitched frame — it is in all rows. Spectracs therefore runs **both**, in that order (§3.7, §3.8).

→ Background: **KB 3 — Robust statistics: median, MAD, sigma clipping and the Tukey biweight**

2.6 The lamp — what a "white" LED spectrum really is

The reference lamp is a Yuji SunWave 6500 K high-CRI white LED. A white LED is not a white light source in the physical sense: it is a **narrow blue pump chip** plus a **broad phosphor** that converts part of that blue into everything else.



The measured reference spectrum. The narrow spike near 474 nm is the blue pump chip itself; the broad hump is phosphor emission. The dip near 580 nm is the phosphor's own gap. This shape is not flat — and it does not need to be, because it appears identically in R and S and divides out.

Two practical consequences follow from that shape, and both were checked on real data:

- **The blue-pump spike is not an artefact of ours.** It is a genuine feature of the lamp at ~473 nm, and it is *steep*. It appears in the reference and in the sample alike, so it cancels: measured across the edge, absorbance is smooth (0.165 → 0.165 → 0.136) even though the raw reference jumps from DN 186 to 228. This is §2.4's first row, visible in the data.
- **The red end is dim.** Beyond ~640 nm the small S-mount optics roll off hard, so that region is unusable. This is a genuine hardware limit, not a processing choice, and it is why the pumpkin evaluation uses two pigment bands rather than three.

→ Background: **KB 6 — White LEDs, colour temperature and colour rendering**

3. The decisions — what we do, and what led us there

Each section below follows the same shape: **the problem**, **what we measured**, **the decision**, and **what it costs**. They are in pipeline order, not chronological order.

3.1 Capture resolution is pinned to 2592×1944

The problem. The camera can deliver several resolutions and will silently substitute the nearest one it supports. The wavelength calibration — the polynomial mapping pixel column $\rightarrow$ nm — is authored at one specific resolution. If capture and calibration disagree, every wavelength is wrong, and *nothing downstream can detect it*: you still get a smooth, plausible-looking spectrum.

What we measured. A production capture was running at 1600×1200 while the stored calibration had been authored at 2592×1944 , because the backend requested 1920×1080 — a mode this camera does not have. The evaluation bands were mapping to the wrong wavelengths, and the Q-band had fallen off the frame entirely. The resolution was proven from physics: the reference peak sits at normalised position 0.69, which is 572 nm (the phosphor peak) only if the frame is 2592 wide.

The decision. Pin capture to 2592×1944 — the highest mode with working exposure control, and the resolution the existing calibration was authored at, so no recalibration was needed. Add a **tripwire**: if the calibration's region of interest does not fit inside the delivered frame, warn once and clamp, so the failure can never again be silent.

What it costs. A per-camera magic number, and slower ~ 1.5 fps bursts at 8 megapixels — which in turn is what made all the exposure timing fragile (§3.5).

3.2 The grey value of a pixel is $\max(R, G, B)$

The problem. The sensor is a colour sensor with a Bayer filter, but a spectrum needs *one* number per pixel. Which one?

The original code used the standard broadcast luminance formula, $qGray = (11 \cdot R + 16 \cdot G + 5 \cdot B) / 32$. That is a **photometric** weighting: it models how bright a colour looks *to the eye*, which deliberately under-weights blue by a factor of three. For a spectrometer this is exactly wrong. At 450 nm the blue-filtered photosites are the ones that actually saw the light; weighting them $5/32$ throws most of the signal away.

What we measured. With $qGray$, the reference trace read ~ 25 in the blue against ~ 115 in the green — yet the raw camera image of that same region was **vividly blue**. The blue channel was strong; the *weighting* was suppressing it.

The decision. The grey value of a pixel is $\max(R, G, B)$ — the brightest channel, i.e. the one whose colour filter matches the light actually falling there. It is *radiometric* rather than photometric.

→ Background: KB 5 — why a Bayer sensor makes $\max$ the physically right choice

Why it is safe. Any homogeneous reduction cancels in $T = S/R$, because at a given wavelength the reference and the sample have the *same colour* and differ only in brightness. So the change does not bias transmission, absorbance or colour at all. What it buys is **signal-to-noise in the blue** — roughly a factor of six — which is what let the Soret band become a usable measurement.

An unexpected bonus, found much later. $\max()$ is an *order statistic*: for any strictly increasing function f , $\max(f(R), f(G), f(B)) = f(\max(R, G, B))$, exactly. That means gamma decoding commutes with the channel combine — it can be applied before or after with identical results. Had we stayed on the weighted sum, it could not (the decode of a sum is not the sum of decodes). A decision taken for signal-to-noise reasons quietly removed a whole class of ordering bug. Medians share this property; averages do not.

3.3 The brightness law: decode with the pure 2.2 power law

Status: designed and verified, not yet implemented. This section records the decision and the evidence behind it.

The problem. §2.2 established that our absorbances are compressed by $\approx 1/2.2$. The obvious fix is to undo the camera's bend before doing any arithmetic. The question is which curve to undo it with — §2.3's variant A or variant B — and whether doing so is safe for the pumpkin verdict.

What we measured. Every Spectracs report PDF embeds the complete workflow as JSON *and* both full-resolution capture frames, so the entire pipeline could be replayed off-line at both the spectrum and the pixel level. The replay reproduces the app bit-for-bit. Two oils measured in 2026, plus the 32-run four-oil set from the capability proof, were re-processed under each candidate decode.

Result 1 — the verdict cannot be moved by a pure power law, at any exponent.

decode model	brown oil ratio	green oil ratio	verdict	perceived chroma	absorbed hue
as-is, no decode	4.0591	5.1826	red / green	33.7 / 32.2	298.2° / 300.0°
pure power $\gamma = 1.8$	4.0591	5.1826	red / green	41.6 / 40.8	298.2° / 300.0°
pure power $\gamma = 2.2$	4.0591	5.1826	red / green	43.9 / 43.5	298.2° / 300.0°
pure power $\gamma = 2.6$	4.0591	5.1826	red / green	45.9 / 45.9	298.2° / 300.0°
true sRGB (with toe)	3.6362	4.7391	red / green	43.5 / 42.7	292.8° / 296.8°

Bit-identical to fifteen significant digits at every exponent — not "close", *identical*. The absorbed colour is equally invariant, because chromaticity is computed after luminance has been divided out, and a uniform scale changes only luminance. **Corollary for a fleet of cameras:** the condition is "a pure power law", not "2.2 exactly", so pigment ratios from cameras with different gammas are already directly comparable today. Only *absolute* absorbance and colour need the decode.

Result 2 — the piecewise sRGB curve is worse for us, and was declined. The toe rescales the Soret band (440–460 nm) by a *sample-dependent* amount, because a browner, darker oil has more of its band below the knee (17.4 % of bins) than a greener one (4.3 %). A non-uniform, sample-dependent rescale of the numerator is precisely what a ratio cannot absorb:

		as-is	pure 2.2	true sRGB
2026 pair	ratio gap	1.123	1.123	1.103 (−1.8 %)
32-run proof set	ratio gap	1.277	1.277	1.156 (−9.5 %)
	within-group scatter (green)	0.134	0.134	0.201 (+50 %)
	separation ÷ noise	10.39	10.39	7.87 (−24 %)

		as-is	pure 2.2	true sRGB
	colour gain (perceived chroma)	29.1	40.6	39.9

The physically more faithful curve costs a quarter of the class separation and buys no measurable colour improvement.

The decision. Decode with `light = (DN/255)^2.2`, applied **per channel, as the very first operation on pixel values** — before the channel combine, before any averaging. The piecewise sRGB EOTF is **deliberately declined**; this is an operational choice over a physical one, recorded so that a later reader does not "correct" it toward the standard and silently lose the separation.

Why "first" matters, by argument rather than by test. The camera has *already* applied its bend before we see anything, so undoing it must precede everything we do. `max()` and medians commute with the decode exactly (§3.2); the two **averages** — Tukey and the sigma-clipped mean — do not, because `mean(x)^γ ≠ mean(x^γ)`. Decoding first makes that gap zero by construction rather than small by measurement. (Measured, had we got it wrong: 0.08–0.20 % on the ratio.)

What it costs, and why we do it anyway. It changes nothing about the verdict — that is the point. The gains are a **+33 to +40 % increase in perceived colour chroma**, physically real absorbance values, and cross-camera comparability of absolute numbers. But the real motive is **closure**: we *know* the camera is non-linear, so as long as the assumption stays in the pipeline, "maybe it's the gamma" remains a suspect for every future anomaly. Removing it retires that suspicion permanently.

One thing must ship in the same change. The colour code clamps absorbance at 3.0 before computing a colour, to stop a near-opaque wavelength from dominating the calculation. Today absorbance peaks at 1.3–1.6 and the clamp never fires. Decoding roughly doubles absorbance, so the clamp would begin cutting real signal. It must be raised to 6.6 at the same time.

→ Background: KB 7 — **From a spectrum to a colour: CIE XYZ, chromaticity, hue and chroma**

3.4 Auto-exposure: a direction-agnostic sweep on the channel peak

The problem. The exposure must be set so that the brightest wavelength is close to, but not at, saturation — and then **locked**, because an exposure that differs between reference and sample is a multiplicative error that does *not* cancel (§2.4).

What we measured — three separate traps.

1. **The camera's exposure control is inverted.** On this ELP module a *lower* exposure value gives a *brighter* image, and above roughly 16 the response flattens entirely. The original algorithm was a bisection that assumed brightness rises with the setting, so it drove straight to the extreme and picked the most clipped image available.
2. **You cannot measure exposure by watching the live video stream.** Frames arrive asynchronously, and a UVC exposure change takes a fixed wall-clock time to take effect. Reading the stream after a change returns frames from *before* it. At 8 megapixels the camera runs at 1–2 fps, which stretched the ramp to ~1.5 s and made every frame-counting scheme fail.
3. **Averaging hides clipping.** A luminance metric averages a clipped green channel together with an unclipped red one and reports a comfortable value while green is pinned at 255.

The decision.

- Metric = **channel peak**: the 99.9th percentile of `max(R, G, B)`, targeted at **245**. Working on the per-channel maximum *guarantees* no channel is saturating, which a luminance average cannot.
- Search = a **direction-agnostic sweep** (coarse ladder, then bisection on the sign of "above/below target"), which makes no assumption about which way the control runs. The lowest exposure value is excluded as a UVC artefact.
- Measurement is **synchronous, inside the video thread**: set the exposure, actively drain and discard frames for a settle period, *then* read. Never off the async stream.
- The chosen exposure is then **locked** and reused for the sample capture.

What it costs. An exposure sweep takes several seconds at 8 megapixels. The upper search bound is still a hard-coded 500, which is not meaningful across camera models — a known open item (§6).

3.5 Settling: wait until the picture stops changing, don't count frames

The problem. After an exposure change the camera does not jump to the new brightness — it *ramps* over roughly a second, and the ramp is neither instant nor of predictable length. Frames captured during the ramp are systematically too dark.

What we measured. The original code discarded a single frame and waited a fixed 1.8 s. That constant was measured for one camera at one resolution, and it silently failed elsewhere. Worse, an early "adaptive" attempt read the *channel peak*, which plateaus as soon as the brightest pixels saturate — while the rest of the frame is still ramping. It declared success early, in the flat part of the curve, and let dim frames through.

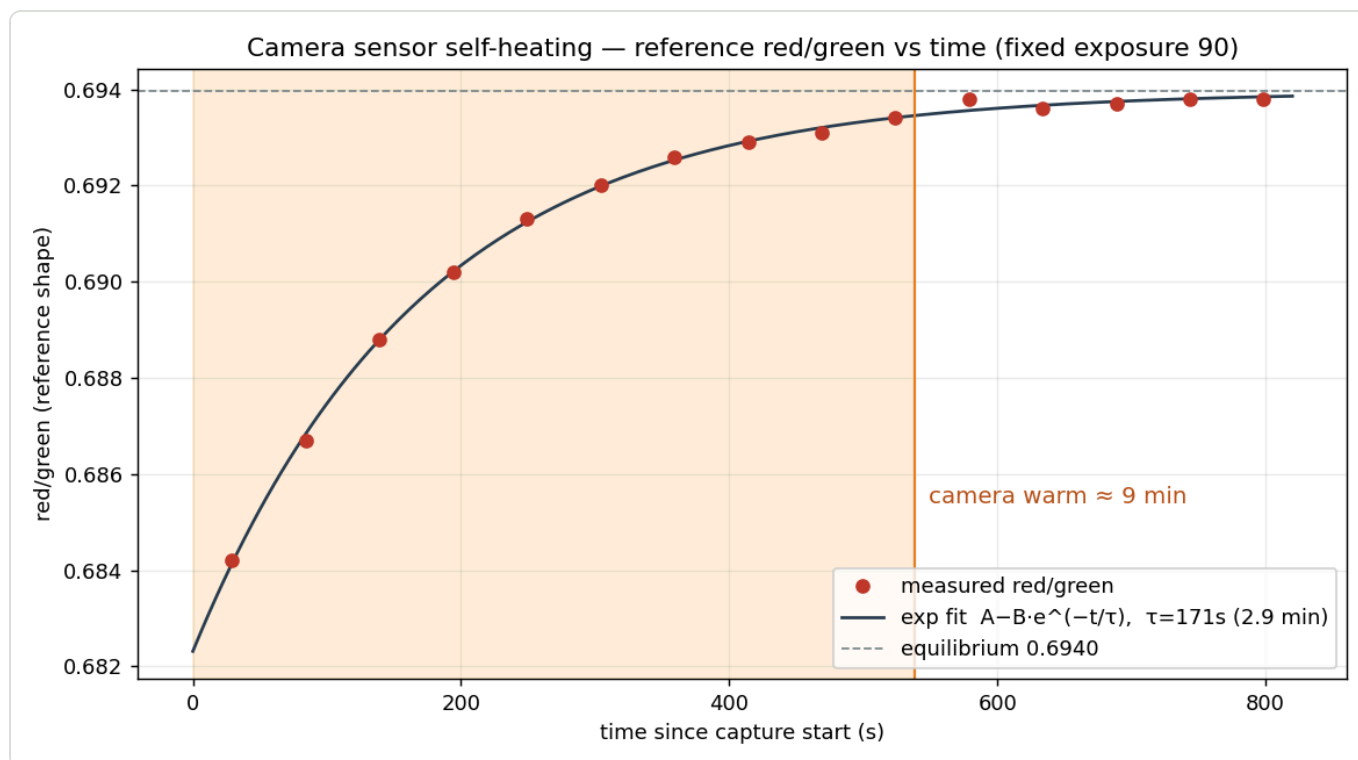
The decision. Settle **adaptively on a quantity that does not plateau**: the mean brightness of the brightest channel over the whole frame. Drain frames in 200 ms chunks until **three consecutive reads agree within 1 %**, with a floor of 1500 ms (never declare success before the known latency plateau) and a ceiling of 4000 ms (a never-settling ramp must not hang the app).

What it costs. Up to four seconds before a burst. In exchange, the timing constant stops being a per-camera magic number.

3.6 Warm-up: the camera heats itself

The problem. Even with the lamp constant, exposure locked and white balance frozen, the reference measured at the start of a session and the reference measured ten minutes later were not the same shape.

What we measured. The channel balance — red divided by green — drifts as a clean single exponential from the moment the camera is opened. Overall brightness stays pinned (the exposure holds it); only the *shape* moves. This is **sensor self-heating**: as the die warms, the per-wavelength sensitivity of the channels drifts, red being the most temperature-sensitive.



Camera sensor self-heating at fixed exposure. The red/green ratio of the reference follows $A - B \cdot e^{(-t/\tau)}$ with a time constant of 171 s, a total shape change of 1.68 %, and full settling after about nine minutes.

quantity	value
time constant τ	171 s (2.9 min)
total shape change	1.68 % (red/green 0.682 $\rightarrow$ 0.694)
90 % / 95 % settled	6.6 min / 8.5 min
indistinguishable from equilibrium	≈ 9 min

Why it bites harder than it looks. The camera is opened and released around each capture. So a session typically opens the camera *cold*, captures the reference in the first seconds — on the steepest part of the curve — and then the sample thirty to sixty seconds later, while it is still warming fast. The reference-to-sample drift is therefore close to its worst case, and it resets every session.

The decision — recorded, not yet enforced. The diagnosis and the curve are established; the remedy (hold the capture button until the shape stabilises, or keep the camera warm across the session) is designed but deliberately not built. In the meantime the operational rule is simply: **let the camera run for a few minutes before the reference**, and capture the reference and the sample close together.

Note that this drift is a *shape* change, not a brightness change, and it therefore does **not** cancel in $T = S/R$ — the reference and sample see different channel balances. It is the largest uncorrected error left in the instrument, at about 1.7 % on the reference shape.

3.7 Spatial reduction: a Tukey biweight down the slit

The problem. Each wavelength column contains many pixel rows — the height of the illuminated slit. The original code read a single centre row, which meant one hot pixel or one dust speck corrupted that

wavelength permanently. But a plain average over the rows is no better: a single stuck-at-255 pixel drags the whole column.

The decision. Reduce each column with a **Tukey biweight** — a robust average that weights each row by how far it sits from the column's median, smoothly dropping to zero weight beyond 6 median absolute deviations. Two refinement passes.

Two details matter as much as the estimator:

- ****Saturated and dead pixels are masked *before* the reduction, and masked per channel**** — saturation is a per-channel fact, so it has to be detected before the channels are combined.
- **The outer 20 % of rows are dropped** (top and bottom). Edge rows bleed the dark border outside the slit and carry the worst optical "smile" distortion. The measurement is broadband, so a generous central band costs nothing.

Why a biweight and not a median. The number of rows is modest, and a median throws away most of the averaging benefit. The biweight keeps the noise reduction of a mean while still discarding a genuine outlier — the right trade at small sample counts.

→ Background: **KB 3 — breakdown point, why MAD is scaled by 1.4826, and when each estimator wins**

3.8 Temporal reduction: reject dim frames, then a sigma-clipped mean

The problem. A burst of 150 frames is averaged to beat down noise. A plain mean is fine until one frame is bad — and worse, until a *group* of frames is bad in the same direction.

What we measured. Reference spectra showed a systematic downward bias that the sample spectra did not have. The cause: the reference capture runs an auto-exposure sweep immediately before its burst (the sample reuses the locked exposure), and the exposure ramp left the **first several frames of the reference burst systematically dim**. A per-wavelength sigma-clip cannot reject that: with a sizeable minority all dim in the same direction, the per-bin spread inflates and the mean is dragged toward them rather than away.

The decision — two stages, because they catch different things.

1. **Reject dim frames as whole frames.** Compute one brightness scalar per frame, then reject frames whose brightness is a median-absolute-deviation outlier against the median frame — with a floor of 2 % so that a pathologically tight cluster of clean frames does not over-reject, and a blatantly dim frame cannot hide behind it. Judging the *whole frame* on a single scalar is what makes the coherent dim group obvious; the 50 % breakdown point of the MAD is what makes it rejectable even when it is large. Symmetric, so a bright flare frame goes too.
2. **Then a sigma-clipped mean per wavelength.** Iteratively drop samples beyond 3σ (σ estimated robustly from the median absolute deviation), then average the survivors — keeping the full $\sqrt{N}$ benefit while a single glitched value is discarded.

Additionally the burst **tops itself up**: it grabs until *N surviving* frames are collected rather than *N grabbed*, so rejection does not quietly reduce the average's depth.

→ Background: **KB 4 — why averaging N frames buys $\sqrt{N}$, and what it cannot buy**

What it costs. Bookkeeping, and a slightly longer burst. It removed a real bias in the reference, which is the one spectrum every result divides by.

3.9 White balance: frozen for measurement, automatic for calibration

The problem. Automatic white balance continuously rescales the R, G and B channels. Between a reference and a sample capture that is a *per-channel multiplicative change*, which breaks the premise of §2.4 — it no longer cancels, because it is not the same factor in both.

What we measured. After the dim-frame work, a residual reference-only anomaly remained, and the lamp was demonstrably constant. The cause was that opening the camera left **automatic white balance and backlight compensation switched on**. They re-converge after every exposure change, and the reference burst — which follows an exposure sweep — catches them mid-convergence.

The decision — split by use case, because the two uses need opposite things.

use	white balance	why
measurement (reference, sample)	fixed at 6500 K	matches the 6500 K lamp; frozen, so it is identical in R and S and cancels exactly
calibration (mercury / europium lines)	automatic	line identification keys on the <i>colour</i> of each emission line; auto-WB is what the historical, working calibration used

Gain and backlight compensation are pinned off in both cases; a fixed frame-buffer depth of one is also set, so a capture never receives a stale queued frame.

What it costs. Two code paths where a naive design would have one — justified because the two uses genuinely have opposite requirements.

3.10 Dark frames: we measured, and they were black

The problem. §2.4's table says any *additive* offset — sensor black level, dark current, stray light — survives the division and biases transmission toward 1, worst exactly where the signal is weakest. Textbook practice is to capture a dark frame and subtract it.

What we measured. 150 dark frames at the longest available integration time (the worst case for dark current), analysed over the region of interest:

quantity	result	interpretation
black level	0.00 % of full scale	no pedestal — true black really is 0
saturated pixels	0 %	—
hot pixels in the region of interest	~10	few, and fixed in place

The decision. Do not implement dark-frame subtraction. The offset it would remove is not measurably present, and the ~10 hot pixels are already handled by the spatial Tukey estimator (§3.7), which is designed for exactly that failure. A per-pixel bad-pixel map was dropped for the same reason.

Why the measurement was worth doing anyway. It converted an assumption into a fact, and it is now load-bearing elsewhere: gamma decoding (§3.3) assumes true black is 0. Had there been a pedestal of 8 or 16 DN, decoding without subtracting it would have swung the pigment ratio by −12 % to +13 % — a *larger* effect than gamma itself. We can decode safely **because** this was measured.

This is the clearest example in the project of Edwin's standing rule: **measure first, then decide**. The measurement's outcome was "do nothing", and that is a perfectly good outcome — it is what makes "do nothing" defensible rather than lazy.

3.11 The wavelength window: 440–630 nm, declared by the plugin

The problem. The camera sees more than the instrument can trust. Below ~440 nm and above ~630 nm the lamp is weak and the small S-mount optics roll off, so those regions contribute noise dressed as data.

The decision. The evaluation plugin **declares** the wavelength window it needs and the bands it will read; the host hard-clamps the captured region of interest to that window. The plugin also asserts at construction time that its window covers every band it declares — so a mismatch fails loudly at start-up rather than silently producing a spectrum with a missing band.

What it costs. Nothing, and it inverts the dependency correctly: the science declares what it needs, the capture layer obeys.

3.12 Sample presentation: keep the signal out of the floor

The problem. This is a *lab* parameter, not a code parameter, but it belongs here because it determines whether the capture chain has anything to work with. If the sample is too concentrated, the transmitted light at the strongest absorption band falls to a handful of DN, where quantisation is coarse and the camera's toe (§2.3) is at its least trustworthy.

What we measured. The 2026 oils are fresher and absorb considerably more than the aged 2023 oils used for the original validation. At the old strength (2 drops in 4 ml, $\approx 1:20$), the sample bottoms out at **DN 5 of 255** at 440 nm — 17 % of the Soret band sitting in the toe. The oil is effectively opaque there.

The decision. Dilute more. Simulated on the measured spectra, the minimum acceptable strength is about **1:27**; the working recipes are **1:30 or 1:33**:

dilution	lowest sample DN at 440 nm (brown / green)	assessment
1:20 (old)	5 / 10	too dark — 17 % of the band in the toe
1:27	12 / 20	the minimum that clears it
1:30 — 6 ml + 2 drops	16 / 25	comfortable
1:33 — 10 ml + 3 drops	21 / 31	comfortable, more headroom

Why this is safe to change mid-series. The pigment ratio moves by ± 0.35 % across every dilution simulated, against an 8.7 % run-to-run spread. That is dilution-invariance doing its job: the recipes are interchangeable, the verdict threshold is unaffected, and older runs stay comparable.

A practical point worth stating plainly. The *transfer* volume does not matter. The measurement sees the solution's concentration and the pot's fixed path length; how many millilitres are poured in is irrelevant as long as the light path is filled. Only the **batch concentration** needs care — which is why the recipe is "prepare a batch at a known concentration, then simply fill the pot", and why a larger batch (more drops, proportionally more solvent) is the cheapest way to improve precision: half a drop out of two is ± 25 %, out of six it is ± 8 %.

And the one instruction that genuinely matters at the bench: **dissolve it properly**. Turbidity is additive (§2.4) and does not cancel — a cloudy sample reads as absorption that is not there.

→ Background: **KB 2 — dilution invariance, and the assumptions it rests on**

3.13 The jar, and the sample's own turbidity

Added 2026-07-31. Everything before this section is about the instrument. This one is about the two things in front of it — the vessel and the liquid — which together turned out to dominate the error budget more than anything downstream of the grating.

The mechanical half — measured, and fixed. Absorbance is a ratio of two captures taken minutes apart with the jar lifted out and put back between them. Any change in how the jar sits changes the path through it, and that enters absorbance as an ****offset and a slope****, not merely a level. A dedicated re-seating probe attributed the bulk of the run-to-run scatter to that one joint.

A rebuilt seat and a marked-line fill protocol then cut it:

	before	after
jar re-seat tilt	2.84 %	1.34 %
metric CV, one fill re-seated	9.7 %	2.9 %

That was the single largest fidelity improvement in the project, and it is worth noting *what kind* of improvement it was: not an algorithm, not a calibration — a printed part and a filling habit.

And a caveat found immediately afterwards. Of that remaining 2.9 %, roughly **58 % is not seating at all** but a slow drift while the freshly-mixed dilution settles. Removing the time trend leaves a true seat-to-seat repeatability of about **1.9 %**. The mechanical work therefore over-delivered on its own terms — and handed the top of the error budget to the chemistry.

The chemical half — measured, not yet fixed. A few drops of oil in isopropanol is **not a solution**. The two are only partially miscible, so what forms is a cloudy dispersion of oil droplets and never-soluble waxes. Scattered light does not reach the detector, so the instrument records it as absorbance: a broad additive floor under the whole curve, the **pedestal**.

It is not small. Measured across eight fills, the pedestal runs **0.7 – 1.9 × the pigment signal in the Q band** — the plinth is bigger than the thing standing on it. And because the metric is a *ratio*, adding the same amount to numerator and denominator drags it toward 1:

true	12.37 / 1.00	= 12.37
measured	(12.37 + 1.59) / (1.00 + 1.59)	= 5.39

That compression is the entire difference between the two pigment-ratio rows in the report, and the reason a baseline correction exists at all.

How much it costs, from an accident. Oils bought in 2023 and measured in 2026 had three years to clarify in the bottle. They carry **half** the pedestal of fresh oils — and they separated the two quality classes about **8× better**. Nobody designed that experiment; the shelf did. It makes sample clarity a **first-class instrument parameter**, and it is currently the largest known lever left.

One consequence for handling. The batch is mixed in a lab glass and a 4 ml aliquot is transferred to the jar — so the transfer is a **sampling step out of a dispersion that is sedimenting the whole time**. How much particulate travels with the aliquot depends on when and from what depth it was drawn. Homogenise before drawing, or clarify after; doing neither leaves a large sample-to-sample variable that is invisible in the finished spectrum.

Background. The miscibility gap, the ouzo effect, why the droplets sediment rather than cream, the solvent-selection table and the vessel constraints are treated properly in

KB_spectroscopy_physics.md §8. The open work is owned by SPEC_capture_quality.md §16.12 and SPEC_capability_proof.md §11.4e–f.

4. What we deliberately did not do

An engineering record is only useful if it also says what was considered and rejected, and why. Everything in this table was examined seriously.

not done	why not
Sensor quantum-efficiency / blackbody-lamp calibration	This is the standard recipe for <i>absolute</i> spectroscopy, where there is no reference. We measure $T = S/R$, and the sensor's response and the lamp's spectrum are common factors in both — they cancel exactly (§2.4). Measuring them would re-solve a problem the reference already solves.
OECF characterisation (mapping the camera's true response curve by stepping exposure)	Would refine the gamma model beyond the standard assumption. Ruled out of scope: the decision that depended on it (§3.3) resolved without it, and the pure power law is verdict-neutral regardless of the true exponent.
The piecewise sRGB EOTF (in favour of the pure 2.2 power law)	More physically faithful, and measurably worse for us: −24 % class separation, no colour gain (§3.3). Declined on evidence, not on convenience.
Dark-frame subtraction	Measured at 0.00 % of full scale — the offset it would remove is not there (§3.10).
A per-pixel bad-pixel map	~10 hot pixels, already absorbed by the spatial Tukey estimator (§3.7).
Spectrum normalisation	$T = S/R$ already self-normalises against the lamp, and exposure is locked across the pair. A further constant scale would cancel in every ratio and bias only absolute absorbance. Documented as a deliberate no-op.
Shifting the Soret band to 450–470 nm (to escape the toe)	Tested on 32 runs: costs 25 % of the discriminating power, because the steep flank being abandoned <i>is</i> the pigment signal. The real fix was the sample, not the band (§3.12).
Trimming the band to 442–460 nm	Genuinely the best-scoring variant (+4 %), but small, and it would force a threshold recalibration. Recorded with its price so it can be picked up if a margin ever tightens.

5. As-built settings

Every value the capture chain depends on, in one place. Values marked † are per-camera and would need revisiting on different hardware.

stage	setting	value	§
camera	capture resolution †	2592 × 1944	3.1
camera	frame buffer depth	1 (always the newest frame)	3.9
camera	analogue gain	0 (off)	3.9
camera	backlight compensation	off	3.9
camera	white balance — measurement	fixed, 6500 K	3.9
camera	white balance — calibration	automatic	3.9
auto-exposure	metric	99.9th percentile of $\max(R, G, B)$	3.4
auto-exposure	target	245 of 255	3.4
auto-exposure	search range †	2 ... 500, direction-agnostic sweep	3.4, 6
settle	chunk / minimum / maximum	200 ms / 1500 ms / 4000 ms	3.5
settle	stability criterion	3 consecutive reads within 1 %	3.5
warm-up	time constant / practical wait	171 s / ≈ 9 min	3.6
pixel	grey value	$\max(R, G, B)$	3.2
pixel	gamma decode (designed)	$(DN/255)^{2.2}$, per channel, first	3.3
spatial	estimator	Tukey biweight, $c = 6 \cdot MAD$, 2 passes	3.7
spatial	rows used	central 60 % (20 % dropped each edge)	3.7
temporal	burst length	150 frames, topped up to 150 survivors	3.8
temporal	dim-frame rejection	MAD , $k = 3$, 2 % scale floor, ≥ 5 frames	3.8
temporal	averaging	sigma-clipped mean, $k = 3$, ≤ 5 passes	3.8
window	wavelength range	440 – 630 nm, plugin-declared	3.11
sample	dilution	1:30 (6 ml + 2 drops) or 1:33 (10 ml + 3 drops)	3.12
sample	solvent	isopropanol, ≥ 99.8 %, fresh bottle	3.13
sample	equilibration after mixing	≈ 15 min	3.13
vessel	jar re-seat tilt, as rebuilt	1.34 % (was 2.84 %)	3.13
lamp	reference source	Yuji SunWave 6500 K, high-CRI white LED	2.6

6. Known open items

Recorded honestly; none of these blocks a measurement today.

- **Sensor warm-up is diagnosed but not handled** (§3.6). At ~1.7 % on the reference shape this is the largest uncorrected error in the instrument. The remedy is designed; a warm-camera habit is the current mitigation.
- **The auto-exposure upper bound of 500 is hard-coded** (§3.4). Exposure units differ wildly between camera models, so this is not portable. It belongs with the other per-sensor values.
- **Gamma linearisation is designed and verified but not implemented** (§3.3). Verdict-neutral, so there is no urgency; the motive is closure.
- **The temporal half of the decode-order check is argued, not measured** (§3.3) — the individual burst frames are not persisted, only their mean. One instrumented bench run would settle it.
- **The new dilution protocol is predicted, not yet confirmed on the rig** (§3.12). One oil measured at both the old and the new strength would verify the invariance directly.
- **Prep noise and instrument noise are now partly separated** (§3.13) — the rebuild took the re-seat spread to 2.9 %, of which ~58 % is settling drift rather than seating. What remains unmeasured is **fill-to-fill** noise: several fills from one batch, which is what gives the quality threshold its first real error bar. Owned by `SPEC_capability_proof.md` §11.4f.
- **Everything about the brown oil post-rebuild is projection** (§3.13). The mechanical work was verified on green only; brown has not been measured since. Pre-rebuild the two classes had *identical* noise, which is why the fix is expected to transfer — but expected is not measured.
- **The sample's turbidity is the largest known remaining lever and nothing has been done about it** (§3.13). Three candidate routes exist — a better solvent (1-butanol), a 0.22 µm filter, or simply a fresher solvent bottle — and none has been tried. Owned by `SPEC_capture_quality.md` §16.12.
- **The far baseline anchor is not what it was documented to be.** The 600–630 nm window was described as oil-quiet; it carries real chlorophyll absorption and a substantial share of the discrimination rests on it. The metric works, but the explanation attached to it has been rewritten — `SPEC_capability_proof.md` §2.1a.
- **The class threshold rests on four oils.** Precision is close to solved; whether the threshold divides good from over-roasted oil *in general* is a panel question no amount of precision can answer. This is the binding constraint on the claim, not the instrument.

Appendix A — Knowledge base

Background for the topics the main text uses but does not stop to explain. Each entry stands on its own and can be read in isolation. Nothing here is specific to Spectracs unless it says so.

KB 1 — The brightness law (gamma) in depth

Where it comes from. Two unrelated facts converged. First, a cathode-ray tube's brightness responds to grid voltage as roughly $\text{voltage}^{2.5}$ — a physical accident of electron optics. Second, human brightness perception is approximately a power law with an exponent near 0.4, i.e. close to the *inverse*. Encoding images with $^{(1/2.2)}$ therefore did two jobs at once: it pre-compensated the CRT, and it happened to store brightness in perceptually uniform steps. CRTs are long gone; the second reason is why the encoding stayed.

Why perceptual coding matters in 8 bits. The eye can distinguish roughly a 1 % change in brightness. Store light linearly in 256 steps and the step from 1 to 2 is a **100 %** jump — grossly visible banding — while the step from 200 to 201 is 0.5 %, finer than anyone can see. Half the code values are wasted at the top and there are far too few at the bottom. Bending the scale before quantising spreads the code values where the

eye needs them. Linear 8-bit imaging is simply not viable; you would need about 12 bits to match what 8 bent bits achieve.

Terminology. The curve has direction-dependent names:

term	meaning
OETF — opto-electronic transfer function	light → stored value. What the camera applies (the <i>encode</i>).
EOTF — electro-optical transfer function	stored value → light. What the display applies (the <i>decode</i>). This is the one we implement in software.
gamma (γ)	loosely, the exponent of either. "Gamma 2.2" conventionally means the <i>decode</i> exponent.
linear light	values proportional to photon count. What physics needs and what a camera does not give you.

The exact sRGB functions (IEC 61966-2-1). With **L** = linear light and **V** = stored value, both normalised to 0...1:

encode (OETF)	$L \leq 0.0031308 \rightarrow V = 12.92 \cdot L$ $L > 0.0031308 \rightarrow V = 1.055 \cdot L^{(1/2.4)} - 0.055$
decode (EOTF)	$V \leq 0.04045 \rightarrow L = V / 12.92$ $V > 0.04045 \rightarrow L = ((V + 0.055) / 1.055)^{2.4}$

Why 2.4 yields "2.2". Three ingredients combine: the exponent 2.4, the scale-and-offset $(V + 0.055) / 1.055$, and the linear segment at the bottom. The offset flattens the curve considerably, and the linear segment flattens it further near black. The *composite* curve tracks a pure $V^{2.2}$ closely across most of the range. Neither 2.2 nor 2.4 describes the whole function exactly — 2.2 describes its behaviour, 2.4 is a construction parameter.

The toe, and why it exists. A pure power curve has infinite slope at the origin in the encode direction: as light approaches zero, an ever smaller change in light produces the same change in stored value. In a quantised, noisy system that is catastrophic — the darkest codes would represent immeasurably small light differences, and decoding would amplify sensor noise without bound. The standard therefore caps the slope at a finite 12.92 and splices a straight line below the knee ($V = 0.04045$, i.e. **DN ≈ 10** of 255). That region is the **toe**. It is the only place the two variants differ meaningfully — and, for a spectrometer measuring a strongly absorbing sample, it is exactly where the interesting measurement can end up.

The classic pitfalls. Any arithmetic on encoded values is wrong unless it is linear-preserving:

- **Averaging or resizing in gamma space** darkens the result — the canonical demonstration is that scaling down a black-and-white checkerboard should give mid-grey and instead gives something much darker.
- **Alpha blending and lighting** in gamma space produce the muddy transitions familiar from older software.
- **Ratios** — our case. S/R on encoded values yields $T_{\text{true}}^{(1/\gamma)}$, not T_{true} . Absorbance comes out compressed by $1/\gamma$. Because that is a *uniform* factor, band ratios are unaffected — which is the single reason our verdict survives untouched.

How to tell whether data is encoded. Photograph a scene with a known 18 % grey card. If the card reads near DN 118 the data is gamma-encoded; if it reads near DN 46 it is linear. Webcams, JPEG, PNG and anything intended for a screen are encoded by default. Machine-vision cameras and camera RAW files are linear.

KB 2 — The Beer-Lambert law and the algebra of ratios

The law. For a dissolved absorbing substance,

$$A(\lambda) = \epsilon(\lambda) \cdot c \cdot l$$

where A is absorbance, $\epsilon(\lambda)$ the molar extinction coefficient — a property of the *substance*, varying with wavelength — c the concentration and l the path length through the liquid. Absorbance relates to what is measured by $A = -\log_{10}(T)$, $T = S/R$.

Why absorbance and not transmission. Absorbance is **linear in concentration**; transmission is exponential. Double the concentration and absorbance doubles, while transmission squares. Every useful quantitative statement is therefore made in absorbance space, and this is why the pipeline converts as early as it can.

The algebra that makes a band ratio invariant. Take two wavelength bands and their mean absorbances A_1 and A_2 . Beer-Lambert gives $A_1 = \epsilon_1 \cdot c \cdot l$ and $A_2 = \epsilon_2 \cdot c \cdot l$, so

$$A_1 / A_2 = \epsilon_1 / \epsilon_2$$

The concentration and the path length **cancel identically** — the ratio depends only on the substance's own extinction spectrum. Everything that multiplies absorbance uniformly disappears the same way: sloppy pipetting, a different pot, a different gamma exponent, an overall gain change. This is a strong result and it is why the pumpkin verdict is a *ratio* rather than a level.

Where it stops holding. The law is an idealisation, and four things break it:

- **Stray light.** Any light reaching the sensor without passing through the sample adds a floor. It caps the measurable absorbance and pulls high readings down.
- **Too much absorbance.** Above roughly $A = 2$ (1 % transmitted) the measurement is dominated by noise and stray light, and the relationship flattens. Practical working range is about $A = 0.1 \dots 1.5$.
- **Scattering / turbidity.** An undissolved or cloudy sample removes light by scattering rather than absorption. It is broadly wavelength-dependent, **additive**, and indistinguishable from real absorption in a single measurement. This is why "dissolve it properly" is the one lab instruction that genuinely matters.
- **Chemical interaction.** At high concentration molecules interact and ϵ itself changes. Not a concern at our dilutions.

Why the ratio still needs the level to be sane. Invariance to concentration is not permission to use any concentration. If the sample is so strong that a band falls into the noise or the camera's toe, the *inputs* to the ratio are corrupted before the algebra begins. Dilution invariance holds across the range where both bands are well measured — which is exactly the range the dilution protocol is chosen to guarantee.

KB 3 — Robust statistics: median, MAD, sigma clipping, Tukey biweight

The problem with the mean. The arithmetic mean has a **breakdown point of zero**: a single arbitrarily wrong value moves it arbitrarily far. The standard deviation is worse — an outlier inflates it, which then makes the outlier look acceptable. Any estimator built on mean-and-SD is therefore self-defeating in the presence of the very outliers it is meant to handle.

Median — breakdown point 50 %. Up to half the values can be arbitrarily wrong before the median moves. The cost is *efficiency*: for clean, normally distributed data the median is only ~64 % as efficient as the mean, i.e. it needs about 1.5× as many samples for the same precision. Excellent protection, poor use of good data.

MAD, and the 1.4826. The robust counterpart of the standard deviation is the **median absolute deviation**, $\text{MAD} = \text{median}(|x_i - \text{median}(x)|)$. For normally distributed data $\text{MAD} \approx 0.6745 \sigma$, so it is conventionally scaled by $1 / 0.6745 = 1.4826$ to make it directly comparable to a standard deviation. That factor is the only reason the number appears in the code; it carries no other meaning.

Sigma clipping. Iteratively: compute a robust centre and scale (median and $\text{MAD} \cdot 1.4826$), discard everything beyond $k \cdot \sigma$ (typically $k = 3$), repeat until nothing more is discarded, then take the plain **mean of the survivors**. This gets the best of both — the median's resistance for the *decision*, the mean's efficiency for the *estimate*. It is the standard reduction in astronomical imaging, and it is what Spectracs uses across frames.

Tukey biweight. Instead of a hard in/out decision, weight each sample by how far it lies from the centre, with the weight falling smoothly to exactly zero beyond a cutoff:

$$u = (x - \text{centre}) / (c \cdot \text{MAD})$$
$$w = (1 - u^2)^2 \quad \text{for } |u| < 1, \quad \text{otherwise } 0 \qquad \text{typically } c = 6$$

Being *redescending* — weight zero, not merely small, far out — it fully discards a genuine outlier while still using the information in every good sample. That makes it the better choice when the sample count is **small**, which is why Spectracs uses it across the ~500 rows of a column while using sigma clipping across the 150 frames.

Judging a group, not a point. All of the above assume outliers are *independent*. If a coherent group of values is wrong in the same direction — for example the first frames of a burst, all dim because the exposure was still ramping — per-point rejection fails: the group inflates the estimated spread, which then accommodates it. The cure is to move up a level and judge each *frame* on a single summary scalar. With MAD's 50 % breakdown point the dim group becomes obvious as a group, even when it is large. This is exactly the reasoning behind §3.8's two-stage design.

KB 4 — Digital numbers, quantisation and noise

DN. A *digital number* is simply the integer a pixel reports — here 0...255. It is not photons, not lux, not anything physical. Two cameras reporting DN 100 need not have received the same light.

Quantisation error. Rounding to an integer costs up to ± 0.5 DN. What matters is that this is an **absolute** error against a **relative** measurement:

signal	quantisation error	relative
DN 200	± 0.5	$\pm 0.25 \%$
DN 50	± 0.5	$\pm 1 \%$
DN 10	± 0.5	$\pm 5 \%$
DN 5	± 0.5	$\pm 10 \%$

A measurement that lands at DN 5 is intrinsically a 10 % measurement, whatever else you do. This is why "keep the signal off the floor" (§3.12) is a fidelity requirement and not merely tidiness.

Photon shot noise. Light arrives as discrete photons, and their count fluctuates with a Poisson distribution: collect N photons and the noise is $\sqrt{N}$. Relative noise is therefore $1/\sqrt{N}$ — bright signals are *relatively* quieter than dim ones, independently of the camera's quality. Combined with quantisation, this means the dark end of a spectrum is noisy for two separate reasons at once.

What averaging buys. Averaging M independent frames reduces random noise by $\sqrt{M}$: 150 frames give roughly a 12× improvement. But averaging is powerless against anything *systematic* —

- a hot pixel is identical in all frames (hence the spatial estimator, §3.7);
- a bias such as a black-level offset averages to exactly itself;
- a group of dim frames biases the mean rather than cancelling (hence §3.8's first stage).

$\sqrt{M}$ also has diminishing returns: going from 150 to 600 frames buys a further factor of two for four times the capture time.

Noise amplification in absorbance. Because $A = -\log_{10}(T)$, a relative error in T becomes an *absolute* error in A : $\delta A \approx 0.434 \cdot \delta T/T$. At high absorbance T is small, so a fixed signal-to-noise ratio in DN produces a growing error in A . This is the quantitative form of the "stay below $A \approx 1.5$ " rule of thumb in KB 2.

KB 5 — How a colour camera sees light

The Bayer mosaic. A sensor's photosites are colour-blind: they count photons across a broad range. Colour comes from a physical filter grid bonded to the sensor — the **Bayer mosaic**, typically one red, one blue and two green filters in each 2×2 cell (green is doubled because human vision is most acute there). So every photosite measures exactly *one* colour channel, and two thirds of the colour information at any location is missing by construction.

Demosaicing. The camera interpolates the missing channels from the neighbours, producing a full-colour image. It is a guess, and it is tuned for photographs. On a spectrum image — where colour changes smoothly and steeply along one axis — the interpolation smears slightly across wavelengths. We cannot avoid it on a UVC camera; it is one reason the instrument's effective resolution is somewhat below its pixel pitch.

Quantum efficiency and why $\max$ is right. Each channel has a broad response curve peaking near 450, 540 and 600 nm, overlapping heavily. In a spectrometer, each *column* of the image carries essentially a single wavelength. At 450 nm the blue-filtered photosites are the ones actually receiving light; the red ones are receiving very little and contributing mostly noise. Taking the **maximum** of the three channels therefore selects the detector that saw the light, wavelength by wavelength, with no calibration required. A luminance average would instead mix the signal with two noisy channels — and a *photometric* average, weighted for the eye, deliberately suppresses blue threefold, which for a spectrometer is precisely the wrong thing.

White balance is per-channel gain. "White balancing" multiplies the three channels by three constants so that a neutral object reads neutral. Harmless in a photograph; hazardous in a ratio, because if the constants change between two exposures they no longer cancel. Hence §3.9.

UVC and its controls. USB Video Class is the standard that lets a webcam work without a driver. It exposes exposure, gain and white balance — but **the units are camera-specific**. An exposure value of 100 means different things on different modules, the direction of the scale is not guaranteed, and the usable range varies. This is why the auto-exposure algorithm must not assume a direction (§3.4) and why the fixed upper bound of 500 is a known portability weakness.

KB 6 — White LEDs, colour temperature and CRI

A white LED is two things. There is no LED that emits white. The overwhelmingly common construction is a **blue pump chip** (typically 450–475 nm) coated with a **phosphor** that absorbs part of that blue and re-emits it as a broad yellow band. Blue that escapes plus down-converted yellow reads as white to the eye.

The characteristic spectrum is therefore a **narrow spike** plus a **broad hump**, with a **dip** between them where neither contributes — visible in the measured reference in §2.6.

Colour temperature (CCT). The temperature in kelvin of the blackbody whose colour the source most closely matches. Lower is warmer/redder (2700 K, incandescent), higher is cooler/bluer (6500 K, daylight). It describes only the *colour*, and says nothing about the spectrum's shape — two lamps of identical CCT can have wildly different spectra. For a spectrometer, CCT is close to irrelevant; shape is everything.

CRI, and why high CRI matters here. The colour rendering index measures how faithfully a source renders reference colours compared with a blackbody of the same CCT. A high-CRI lamp is one whose spectrum has **no large gaps** — which is exactly the property a reference source needs. Any wavelength where the lamp is weak is a wavelength where $T = S/R$ divides a small number by a small number and the result is noisy. A cheap low-CRI LED has a deep hole around 480 nm and a weak red end; both would show up as noise bands in the transmission spectrum. Choosing a high-CRI lamp is a signal-to-noise decision, not an aesthetic one.

Why the lamp's own shape does not otherwise matter. The spike, the hump and the dip all appear identically in the reference and in the sample, so they divide out exactly (§2.4). The lamp does not need to be flat — it needs to be **non-zero everywhere and stable in time**. Stability is why the lamp is left permanently on.

KB 7 — From a spectrum to a colour

The CIE 1931 observer. Human colour vision has three cone types, so any spectrum collapses to three numbers. The standard observer defines three matching functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$, and a spectrum $P(\lambda)$ becomes

$$X = \int P(\lambda) \cdot \bar{x}(\lambda) \, d\lambda \qquad Y = \int P(\lambda) \cdot \bar{y}(\lambda) \, d\lambda \qquad Z = \int P(\lambda) \cdot \bar{z}(\lambda) \, d\lambda$$

Y is defined to coincide with luminance. XYZ is a device-independent description of colour: two different spectra with the same XYZ look identical, which is the whole phenomenon of metamerism.

Chromaticity — dropping the brightness. Normalising removes the intensity dimension:

$$x = X / (X+Y+Z) \qquad y = Y / (X+Y+Z)$$

The pair (x, y) is the colour *without* the brightness. This step has a consequence that matters to us: **scaling a spectrum does not change its chromaticity**. So a colour computed from *absorbance* is dilution-invariant for exactly the same algebraic reason a band ratio is (KB 2) — and it is invariant to the gamma exponent too (§3.3).

Hue, chroma, lightness. Converting to HSL gives more intuitive coordinates. Two cautions:

- **Saturation misbehaves at the extremes.** A near-white colour can report ~100 % saturation while being visually almost grey. **Chroma** — computed as $(1 - |2L - 1|) \cdot S$ — stays small there and is the honest measure of colourfulness. Spectracs uses chroma, never raw saturation, when deciding whether a colour is meaningful.
- **Gamut clamping distorts hue.** Colours outside what sRGB can represent get clamped, and the clamping can shift the reported hue substantially. A deeply saturated absorbance colour sits far outside the sRGB gamut, so a hue read straight off a clamped conversion can be an artefact rather than a measurement.

Transmitted colour versus absorbed colour. They are different objects. The colour of the *transmitted* light is what the eye sees and depends on concentration (double the oil and it looks darker and more saturated). The colour derived from the *absorbance* spectrum describes what the substance takes out of the light; it is dilution-invariant, and it is roughly the complement of the perceived colour. The colorimetrically correct way to convert between them is to reflect the chromaticity through the white point — $2 \cdot \text{white} - \text{colour}$, the "mixing-to-white" opposite — which lands far closer to the true perceived hue than naively adding 180° to an HSL hue.

Appendix B — Where the evidence lives

This document summarises; the specifications hold the derivations, the raw numbers and the decision history.

topic	specification
the whole capture chain, all topics	docs/SPEC_capture_quality.md
grey value $\max(R,G,B)$	SPEC_capture_quality.md §15
gamma: design, and the measured verification	SPEC_capture_quality.md §17, §17.5
auto-exposure saga and tuning	SPEC_capture_quality.md §14.5 – §14.7
dim-frame rejection, adaptive settle	SPEC_capture_quality.md §14.8
white-balance split	SPEC_capture_quality.md §14.8
sensor self-heating	SPEC_capture_quality.md §16
dark-frame measurement	SPEC_capture_quality.md §4, §5
resolution pin and tripwire	SPEC_capture_quality.md §4.9
spatial / temporal reduction	SPEC_capture_quality.md §6
wavelength window	SPEC_capture_quality.md §9
pigment bands, band-placement re-test	SPEC_pumpkin_peak_ratio_eval.md §1b, §1b.3
dilution protocol, lab procedure	SPEC_capability_proof.md §7.3
what was actually run, dated	LAB_DIARY_capability_proof.md
the quality verdict built on all this	SPEC_roast_ampel.md
physics background	docs/KB_spectroscopy_physics.md

Appendix C — How this document is produced

The PDF is generated; it is never hand-edited.

```
source of truth  docs/DOC_capture_fidelity.md          (this file, markdown)
generator       docs/tools/build_capture_fidelity_pdf.py
output         ../spectracs-docs/internal/Spectracs_CaptureFidelity.pdf

python3 docs/tools/build_capture_fidelity_pdf.py
python3 docs/tools/build_capture_fidelity_pdf.py --out /tmp/preview.pdf --html
```

To update it: edit the markdown, re-run the generator, commit the markdown *and* the PDF. The generator converts a small markdown subset (headings, lists, tables, code blocks, block quotes, images, `<!--T0C-->` and `<!--PAGEBREAK-->` markers) into a print-styled page and drives headless Chrome to render it — so the only external dependency is Chrome, which the sibling `build_capability_status_pdf.py` already required.

Figures are embedded from `spectracs-references/` as data URIs, so the PDF is self-contained and can be sent to the lab as a single file.

Spectracs · internal engineering documentation. Generated from `docs/D0C_capture_fidelity.md` by `docs/tools/build_capture_fidelity_pdf.py` — edit the markdown and re-run; never hand-edit this PDF.